

## Curated Research Articles

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- **Cathode-Dependent Electrochemical–Mechanical Coupling Governing Edge Lithium Plating and Interfacial Void Heterogeneity of Lithium-Ion Batteries** — score: 1.000 The study reveals that edge lithium plating in lithium-ion batteries, particularly with LiFePO<sub>4</sub> (LFP) cells, is influenced by the mechanical and electrochemical interactions between electrodes during cycling. Specifically, the asynchronous expansion and contraction of LFP and graphite electrodes create mechanical stress heterogeneity, leading to uneven lithium plating and void formation at the edges of pouch cells. It also suggests a strategy for enhancing battery longevity through uniform pressure regulation to mitigate this issue. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Spatial-Interfacial Dual Confinement in Phase-Separated Polymer Electrolytes for Cryogenic Lithium Metal Batteries** — score: 1.000 The article discusses a novel quasi-solid-state polymer electrolyte (QSPE) designed for lithium metal batteries that operates effectively in low-temperature conditions. By utilizing a phase-separated structure with a dual-confinement mechanism, the electrolyte enhances ion transport while stabilizing the lithium interface, resulting in improved cycling stability at temperatures as low as  $-20^{\circ}\text{C}$  and an innovative approach for cryo-tolerant battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic DFOB—/NO<sub>3</sub>—-Regulated Solvation Enables Safe High-Voltage Lithium Metal Batteries** — score: 1.000 The article discusses the development of a dual-anion-regulated electrolyte that combines lithium difluoro(oxalato)borate and lithium nitrate in a nonflammable solvent, triethyl phosphate. This innovative design alters the Li<sup>+</sup> solvation structure, leading to enhanced thermal stability and improved solid-electrolyte interphases that prevent dendrite formation and support high-voltage operation, ultimately resulting in safer and longer-lasting lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reversible Secondary-Solvation-Shell Regulation Enables Dendrite-Free Zinc Anodes via a Polarizable Two-Segment Additive** — score: 1.000 The article presents a novel screening strategy for electrolyte additives that enhance the stability of zinc anodes in batteries by utilizing secondary-solvation-shell interactions. It identifies isopropyl -D-1-thiogalactopyranoside (IPTG) as a key additive that improves zinc ion solvation dynamics and promotes dendrite-free zinc deposition, achieving remarkable battery performance, including over 5,000 hours of stable operation and nearly perfect Coulombic efficiency over extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electrochemical and Thermal Interplay in Blended Cathodes for All Solid-State Batteries** — score: 1.000 This study focuses on blended cathodes composed of LFP and NMC for all-solid-state batteries, showcasing improved rate performance and thermal stability. Through operando X-ray diffraction, the research uncovers charge distribution anomalies and thermal modeling indicates that these blended cathodes produce less heat than pure NMC, suggesting a better balance of efficiency and safety for future battery designs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Cation–Anion Engineering Enables Yttrium Chlorobromide Superionic Conductors for High-Performance All-Solid-State Lithium Metal Batteries** — score: 1.000 The article presents a new halide solid electrolyte, Li<sub>2.5</sub>Y<sub>0.75</sub>W<sub>0.25</sub>Cl<sub>3.5</sub>Br<sub>2.5</sub>, that utilizes combined cation and anion doping to enhance lithium ion conductivity and stability. With a remarkable room-temperature ionic conductivity of 6.38 mS cm<sup>−1</sup> and a stable interface achieved through an in situ formed WO<sub>3</sub> layer, this electrolyte enables long-lasting performance in all-solid-state lithium metal batteries, maintaining high discharge capacity over 500 cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Atomic-Scale Tracing of Lithium Trapped in Copper Current Collectors** — score: 1.000 This study utilizes atomic-scale imaging to investigate the diffusion and trapping of lithium in copper current collectors during cycling of lithium-metal batteries. It finds that lithium accumulates at grain boundaries and junctions after just one cycle, leading to nanocrystalline surface changes and defects

that further entrap lithium, oxygen, and carbon, potentially degrading the copper current collector's performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Minimized Li<sup>+</sup> Coordinating Effect Enables 4.7 V-Class Li<sub>1.2</sub>Mn<sub>0.53</sub>Ni<sub>0.20</sub>Co<sub>0.07</sub>O<sub>2</sub>-Li Batteries** — score: 1.000 This article presents a novel fluorinated ester electrolyte that reduces Li<sup>+</sup> coordination, enhancing the solvation chemistry and forming protective lithium fluoride (LiF) interphases on both the lithium metal and LRMO cathodes. This innovation mitigates issues like dendrite growth and oxygen release, enabling high-performance 4.7 V-class lithium metal batteries with impressive capacity retention even under challenging cycling and high-temperature conditions, achieving a 98.5% coulombic efficiency in tests. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Heavy Lanthanide-Induced Epitaxial Rock-Salt Shells Enable Chemo-Mechanically Robust Sodium-Ion Cathodes** — score: 1.000 The study presents a novel approach to enhancing the stability and performance of sodium-ion batteries by constructing a protective epitaxial rock-salt shell around O3-type cathodes, using heavy late lanthanides to exploit ionic radius mismatches. This innovative design effectively mitigates issues like intergranular microcracking and anionic redox reactions, resulting in minimal voltage decay and significantly improved capacity retention over cycles, thereby paving the way for durable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecular Interfacial Regulators Enable Stable Sulfide Electrolytes for High-Performance All-Solid-State Batteries** — score: 1.000 The study introduces molecular surface regulators, particularly fluoropyridines, that enhance the stability of sulfide electrolyte interfaces in all-solid-state batteries. By effectively suppressing interfacial decomposition during both high voltage and lithium metal operation, the modified electrolyte demonstrates improved performance metrics, including higher critical current density and Coulombic efficiency, resulting in better capacity retention over multiple cycles compared to standard configurations. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electrolyte Confinement Enables a Continuous Transition From Battery-Like to Capacitor-Like Na<sup>+</sup> Storage via Solvent Co-Intercalation** — score: 1.000 This study explores how the choice of electrolyte affects sodium-ion storage behavior in layered titanates, revealing that diglyme electrolyte promotes a continuous transition from battery-like to capacitor-like performance due to solvent co-intercalation and reduced impedance. In contrast, carbonate electrolytes lead to resistive interphases that hinder performance, underscoring the importance of electrolyte solvation and nanoconfinement in enhancing sodium storage kinetics for high-power applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Disorder-Driven Ionic Mobility Edge and Localization-Driven Dendrite Formation in Solid Electrolytes** — score: 1.000 The article presents a novel framework explaining dendrite formation in solid electrolytes, attributing it to disorder-driven localization of ionic transport. It details how migration-barrier disorder impacts the ionic conduction pathways and leads to the emergence of fractal dendrites under electrical bias, proposing that controlling the distribution and mean of migration barriers is crucial for enhancing solid electrolyte stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Additive-Driven Regulating Solvation Structure Toward Stable High-Voltage Lithium Metal Batteries** — score: 1.000 This article explores the innovative use of VC, FEC, and LiBOB additives in carbonate electrolytes, highlighting their role in altering lithium ion solvation structure. By promoting a more stable interphase and mitigating issues like electrolyte oxidation and lithium dendrite formation, the additives enhance the performance and longevity of high-voltage lithium-metal batteries, achieving significant cycle stability and capacity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Atomic-Layer-Deposited ZnF<sub>2</sub> on Aluminum Enabling In Situ NaF/Na-Zn Interphase for Robust Anode-Free Sodium Batteries** — score: 1.000 The article discusses a novel method to enhance the performance of anode-free sodium batteries by applying an atomic-layer-deposited (ALD)

ZnF<sub>2</sub> layer on aluminum current collectors. This coating transforms into a NaF/Na–Zn interphase during initial charge, promoting uniform sodium deposition and significant cycling stability, leading to a Coulombic efficiency of approximately 99.76% under high-demand conditions. This approach offers a scalable solution for advanced sodium-based energy storage technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **A Large Scale Multi-Modal Workflow for Battery Characterization: From Concept to Implementation** — score: 1.000 The article presents a comprehensive multimodal workflow for battery characterization developed through collaboration among fifteen European labs. By integrating various experimental techniques and datasets, the study explores how factors like electrolyte composition affect electrode properties and cell performance over time, revealing that scientific conclusions are highly dependent on chosen observables and techniques. This innovative approach establishes a framework for more holistic battery research, emphasizing the complex interplay between measurement methods, material behavior, and interpretation of results. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Precursor-Regulated Stacking Faults Enable High-Voltage Stability for P2-Type Layered-Oxide Sodium-Ion Cathodes** — score: 0.900 The study reveals that the choice of precursor chemistry significantly influences the structural stability of P2-type sodium-ion cathodes, specifically Na<sub>0.67</sub>Ni<sub>0.33</sub>Mn<sub>0.67</sub>O<sub>2</sub>. By utilizing carbonate precursors, the resulting cathodes incorporate O3-type stacking faults that enhance high-voltage performance by acting as strain buffers, thus improving capacity retention and mitigating structural degradation. In contrast, hydroxide-derived cathodes suffer from instability due to a more ordered structure that leads to detrimental phase transitions during cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Anion–Cation Dual-Site Co-Doping Unlocking NASICON Na<sub>4</sub>VMn(PO<sub>4</sub>)<sub>3</sub> Cathodes Toward Advanced Sodium–Ion Batteries** — score: 0.900 The article presents a novel dual-site co-doping approach using Ti<sup>4+</sup> and Cl<sup>–</sup> in Na<sub>4</sub>VMn(PO<sub>4</sub>)<sub>3</sub> cathodes for sodium-ion batteries, which effectively mitigates Jahn-Teller distortion and lowers Na<sup>+</sup> migration barriers. This innovation significantly boosts the structural stability and electrochemical performance of the material, resulting in enhanced capacity and cycling stability, thereby paving the way for advanced sodium-ion battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Machine Learning for Zinc-Based Energy Storage Devices: From Data-Driven Design to Mechanistic Insights** — score: 0.900 The article reviews the application of machine learning in optimizing zinc-based energy storage systems by facilitating feature selection and elucidating mechanisms that influence performance. It highlights the role of ML in understanding ion transport, interfacial chemistry, and degradation processes, emphasizing the construction of meaningful descriptors that connect material properties with electrochemical behavior. The review also points to future prospects for integrating data-driven modeling with electrochemical theory to enhance the design of these energy storage devices. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multiscale Characterizations Boosting Understanding of Gassing Behavior in Rechargeable Batteries: Progress and Outlook** — score: 0.800 This article discusses advancements in multiscale characterization techniques that enhance the understanding of gassing behavior in rechargeable batteries. It highlights the progress made in this field and outlines future directions for research, emphasizing the importance of these insights for improving battery performance and safety. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Investigation Into the Electrochemical Performance of Micron- and Nanosized Tin in Diglyme and Carbonate Electrolytes in Sodium Ion Batteries (Adv. Energy Mater. 27/2026)** — score: 0.800 The study by John T. S. Irvine and colleagues examines how the size of tin particles affects the electrochemical performance of sodium ion batteries in different electrolytes. They discovered that diglyme electrolytes help maintain the electrode’s structural integrity despite significant volume changes, whereas carbonate electrolytes lead to interphase instability and faster capacity loss due to catalytic decomposition on the tin surface. This research highlights the critical relationship between particle size, electrolyte composition, and long-term battery performance. Journal:

- **Designing Polysulfide-Resistant Solid Electrolyte Interphase for Long-Cycling Lithium–Sulfur Batteries** — score: 0.800 The study focuses on the development of a lithium fluoride (LiF)-rich solid electrolyte interphase (SEI) to combat the corrosion of lithium polysulfides in lithium-sulfur (Li-S) batteries. This innovation effectively increases the cycling lifespan of the batteries, achieving significant improvements in both coin and pouch cells, and highlights the importance of LiF in enhancing battery stability and efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Li Chemical Potential Driven Phase Stability of NCM–Oxide Electrolyte Co-Sintered Composite Cathodes** — score: 0.800 This study investigates the phase stability of NCM811 cathodes when co-sintered with various oxide electrolytes, finding that Ta-doped LLZTO shows superior compatibility compared to LATP and LLTO due to its higher Li chemical potential. The research highlights the Li chemical potential as a critical metric for assessing the stability of these materials, offering guidance for the design of oxide electrolytes in all-solid-state batteries (ASSBs) and suggesting optimization strategies for their practical development. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Engineering Integrated Tandem-Structured Hosts to Achieve High-Loading and Ultra-Cycling Aqueous Zn–I<sub>2</sub> Batteries** — score: 0.800 The study presents a novel integrated tandem-structured host, combining ultrafine silver nanoparticles and N-doped mesoporous carbon fibers, designed to enhance the performance of zinc-iodine batteries. This innovative host effectively immobilizes iodine species, achieving high loading capacity and impressive cycling stability, with the Ag@CMFs/I<sub>2</sub> cathode demonstrating a remarkable 80.7% capacity retention after 92,000 cycles. Additionally, the battery maintains a high Coulombic efficiency and delivers substantial areal capacity, paving the way for practical applications in energy storage. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Anion-Cation Co-Tailored High-Flash-Point Phosphate-based Electrolyte Enables High-Voltage, Wide-Temperature, Non-Flammable Sodium Metal Batteries via an Inorganic-Rich Solid Electrolyte Interphase** — score: 0.800 The article introduces a novel phosphate-based electrolyte designed for sodium metal batteries, addressing key challenges like interface stability and dendrite growth. Through synergistic regulation of anions and cations, the electrolyte develops robust electrode interfaces and enables impressive cycling performance, including 80% capacity retention over 3000 cycles at high voltages and extreme temperatures, while remaining non-flammable. This approach could enhance the compatibility of metal anodes across various battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Abnormal Capacity Rise During Early Stage of Calendar and Cycling Aging for Li-Ion Batteries: A Review** — score: 0.800 The article reviews the phenomenon of abnormal capacity rise observed in Li-ion batteries during early aging, challenging the traditional view of continuous capacity decline. It explores mechanisms such as the anode overhang effect and surface film dynamics, which contribute to this non-monotonic aging behavior. The insights gained could lead to new strategies for enhancing the lifespan of batteries across various applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Aqueous Synthesis of Na<sub>3-2x</sub>Sb<sub>1-x</sub>W<sub>x</sub>S<sub>4-x</sub>I<sub>x</sub> Chalcogenide Solid-State Electrolytes With Ultrahigh Ionic Conductivity** — score: 0.800 The article discusses a new aqueous synthesis method for creating W/I co-doped Na<sub>2.8</sub>Sb<sub>0.9</sub>W<sub>0.1</sub>S<sub>3.9</sub>I<sub>0.1</sub> chalcogenide solid-state electrolytes, which achieve an impressive ionic conductivity of 10.3 mS cm<sup>-1</sup>. The research uncovers that oxygen, not moisture, primarily drives the chemical degradation process, challenging existing assumptions and highlighting the potential for developing high-performance all-solid-state sodium-ion batteries with excellent cycling stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Harvesting Mn<sup>2+</sup> by Encapsulated MnO<sub>x</sub> With sp<sup>2</sup>-Carbon to Enhance the Mn<sup>2+</sup>/MnO<sub>2</sub> Conversion for High-Capacity Zn-Ion Batteries** — score: 0.800 This study presents a novel MnO<sub>x</sub> cathode encapsulated in an in situ formed sp<sup>2</sup>-carbon shell, enhancing

the electrochemical performance of aqueous zinc-ion batteries by effectively preventing  $\text{Mn}^{2+}$  dissolution and facilitating reversible  $\text{Mn}^{2+}/\text{MnO}_2$  conversion. The cathode achieves an impressive capacity of 438.2 mAh g<sup>-1</sup> with a 91.1% capacity retention over 5000 cycles, demonstrating a promising strategy for developing high-capacity, long-lasting Mn-based cathodes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Gradient Interfacial Layer Design for Zinc Metal Anode: Breaking the Trade-Off Between Zincophilicity and Zincophobicity** — score: 0.800 The study introduces an innovative artificial interface layer for zinc metal anodes, designed with a gradient of zincophilicity and zincophobicity to enhance both thermodynamic stability and ion deposition kinetics. This polydopamine-poly(sulfobetaine methacrylate) layer dramatically improves cycling performance in symmetric cells and zinc-ion hybrid capacitors, demonstrating over 4200 hours of stable operation and significant capacity retention over extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Beyond Boundaries: Redox Mediators as Dynamic Charge Carriers and Catalytic Hubs for Efficient Sulfur Conversion in Solid-State Lithium–Sulfur Batteries** — score: 0.800 This review analyzes the role of redox mediators (RMs) in enhancing the performance of all-solid-state lithium–sulfur batteries (ASSLSBs) by addressing the limitations related to sulfur redox kinetics and interfacial challenges. It evaluates different sulfur conversion pathways and the relationship between RM molecular structures and their electrochemical efficiency, identifying key challenges and proposing design strategies for future RM development to optimize battery performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **The Role of Post-Processing in 3D Printed Batteries** — score: 0.800 The article emphasizes the significance of post-processing techniques, such as sintering and electrodeposition, in enhancing the performance of 3D printed batteries, which often lack the necessary properties for practical use in their as-printed form. It offers a comprehensive review of the various post-processing strategies and their effects on optimizing battery microstructures and interfaces, while also outlining future research directions to further improve 3D printed battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Progress and Prospect of Electrolyte Engineering Strategies for Wide-Temperature Lithium–Sulfur Batteries** — score: 0.800 This review addresses the challenges and advancements in the design of electrolytes for lithium-sulfur (Li-S) batteries intended for wide-temperature applications. It analyzes the mechanisms and temperature-related issues of Li-S batteries, classifying recent electrolyte engineering strategies into weakly, highly, and moderately solvating systems, while also suggesting future research directions to enhance electrolyte performance under varying thermal conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Polyrotaxane-Based Polymer Electrolytes With Selective  $\text{Na}^+$  Channels for Stable, Long-Cycling Sodium Metal Batteries** — score: 0.800 This study presents a novel crosslinked polyrotaxane-based polymer electrolyte designed for sodium metal batteries, featuring selective pathways for  $\text{Na}^+$  transport and confined anions. The electrolyte demonstrates impressive ionic conductivity (2.1 mS cm<sup>-1</sup>) and a high sodium ion transference number (0.88), contributing to 2500 cycles of stable Na plating/stripping with 95.9% capacity retention, addressing key challenges of conventional polymer electrolytes in SMB applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **0.71% Lattice-Mismatched Intergrowth Achieves 2000-Cycle Stability in O3-Type Na-Layered Oxide Cathodes** — score: 0.800 This study presents a novel sodium-based layered oxide cathode featuring a unique lattice-matched intergrowth structure that combines the O3 and O 3 phases with minimal lattice mismatch (0.71%). This design enhances cycling stability, yielding a remarkable capacity retention of 79.7% after 2000 cycles, while effectively reducing volume expansion and transition metal dissolution, thus advancing the potential for high-performance sodium-ion batteries in large-scale energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Triple Active Sites in Prussian Blue Analogues for High-Capacity Aqueous and Seawater Ammonium Ion Batteries (Adv. Energy Mater. 25/2026)** — score: 0.800 The study presents a Co/Mn co-doped Prussian blue analogue designed to enhance the performance of aqueous ammonium-ion batteries. By utilizing a synergistic co-doping approach, the research activates multiple redox-active sites, improves structural stability, and facilitates efficient energy storage in seawater electrolytes, paving the way for sustainable energy solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Asymmetric Multi-Site Ion Exchange in Porous Carbon Electrodes (Adv. Energy Mater. 25/2026)** — score: 0.800 The study explores ion transport in hierarchical porous carbon electrodes using 2D EXSY NMR analysis to depict asymmetric multi-site ion exchange. It uncovers distinct kinetic behavior, highlighting the variations in ion mobility depending on proximity to the surface and the complexity of pore networks, thus correlating carbon structure with its electrochemical performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Efficient Polysulfides Adsorption and Catalytic Conversion on Germanium Dioxide for High-Performance Lithium-Sulfur Batteries** — score: 0.800 The article presents the development of GeO<sub>2</sub>@S/C composites designed to enhance the performance of lithium-sulfur batteries by addressing the challenges of sluggish redox kinetics and polysulfide shuttle effects. The optimized 5% GeO<sub>2</sub>@S/C cathode demonstrates significant improvements in capacity, rate capability, and cycling stability, attributed to enhanced electron transfer and efficient lithium polysulfide conversion. Journal: *Wiley: Small Methods: Table of Contents*
- **In Situ Zn-Doped Prussian Blue Analogues as High-Rate Cathodes for Wide-Temperature Sodium-Ion Batteries** — score: 0.800 The study presents an innovative approach utilizing in situ Zn doping in sodium iron hexacyanoferrate (FeHCF) to enhance the performance of sodium-ion battery cathodes. The resultant Zn-FeHCF-2 exhibits superior electrochemical characteristics, including high capacity retention (96.5%) after 6000 cycles at  $-20^{\circ}\text{C}$ , attributed to reduced lattice strain and improved sodium-ion diffusion. Journal: *Wiley: Small Methods: Table of Contents*
- **Synergistic Regulation of Water Activity and Ion Transport by an In Situ-Grown MOF/Bacterial Cellulose Separator for High-Performance Zn Anodes** — score: 0.800 The study presents an innovative separator made from an in situ-grown metal-organic framework (MOF) within a bacterial cellulose hydrogel, addressing the challenges of dendritic growth and water-related side reactions in aqueous zinc-ion batteries (AZIBs). This hybrid separator enhances mechanical strength and facilitates efficient ion transport, thereby enabling long-lasting, stable zinc anodes and improving overall battery performance significantly. Journal: *Wiley: Small Methods: Table of Contents*
- **Weakening Ion Transport Anisotropy Activates Superionic Conduction in Li<sub>2</sub>ZrCl<sub>6</sub>** — score: 0.800 The study reveals that reducing ion transport anisotropy through synergistic In<sup>3+</sup>/F<sup>-</sup> co-doping in Li<sub>2</sub>ZrCl<sub>6</sub> significantly enhances Li-ion diffusion, leading to superionic conduction in the material. The optimized electrolyte exhibits impressive ionic conductivity ( $1.03 \times 10^{-3} \text{ S cm}^{-1}$ ) and excellent performance in all-solid-state batteries, including high capacity retention and stability over extensive cycling, thus positioning it as a promising candidate for advanced battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Fluoride ions as charge carriers in electrochemical energy storage** — score: 0.700 The article presents a review on the potential use of fluoride ions as charge carriers in electrochemical energy storage systems, particularly as alternatives to conventional lithium-ion batteries. It covers key design principles for electrodes, material innovations, degradation mechanisms, and identifies important research directions necessary for advancing fluoride-anion technology. Journal: *Nature Materials*
- **Toward Next Generation Lithium-Ion Battery Recycling: From Metal Extraction to In Situ Cell Repair** — score: 0.700 The article outlines a comprehensive, diagnosis-driven framework for recycling spent lithium-ion batteries, evolving from traditional metal extraction methods to more advanced techniques like materials regeneration and in situ cell repair. This innovative approach

emphasizes the preservation of functional value and battery health, proposing a structured pathway that integrates nondestructive analysis and intelligent evaluation to enhance material recovery and recycling efficiency for the next generation of electric vehicles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **FeOF Nanocomposites With Tunable Graphitic Carbon Shells as High-Performance Conversion-Type Cathode for Potassium Ion Batteries** — score: 0.700 This study presents a novel approach to synthesizing FeOF nanocomposites with tunable graphitic carbon shells using a chelation-assisted polymerization method, which enhances the material’s conductivity and stability. The resulting FeOF@GC-1 electrode shows promising electrochemical performance, achieving a reversible capacity of 132.7 mAh g<sup>-1</sup> after 200 cycles, and supports a full cell with a notable energy density of 181 Wh kg<sup>-1</sup>, marking advancements in conversion-type cathodes for potassium-ion batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **Building chemical stability with mechanics** — score: 0.600 A recent study highlights the importance of stack pressure in lithium-ion battery performance, demonstrating that precise control of this parameter can significantly enhance chemical stability. The research identifies an optimal pressure range that effectively mitigates degradation pathways, leading to a doubling of the battery’s lifespan. Journal: *Nature Energy*
- **Conformal Li3PO4-Coating Synergized Epitaxial Mn-Doping Consolidating Chemical and Structural Stability of Ni-Rich LiNi0.9Co0.1O2 Cathode** — score: 0.600 The article discusses a method to improve the chemical and structural stability of Ni-rich LiNi0.9Co0.1O2 cathodes by applying a conformal coating of Li3PO4 and incorporating Mn-doping. This synergistic approach enhances both performance and longevity of the cathode material, addressing common challenges in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Recent Advances in Electrolyte Additives for Zinc Flow Batteries: Mechanisms, Applications, and Perspectives** — score: 0.500 This article reviews the latest developments in electrolyte additives for zinc flow batteries, focusing on their operational mechanisms and applications. The authors discuss how these advancements can enhance battery performance and provide insights into future research directions in the field of energy storage materials. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Harnessing Selenium Vacancies in Bi2Se3 Cathode via Heterovalent Sn-doping Driven Charge-Compensation and Bond-Reinforcement for Aqueous Zinc-ion Batteries** — score: 0.500 The article discusses a novel approach to enhancing the performance of Bi2Se3 cathodes in aqueous zinc-ion batteries through heterovalent Sn-doping. This process is aimed at utilizing selenium vacancies for charge compensation and improving bond stability, ultimately leading to more efficient energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Suppression of Metallic Lithium in the Solid Electrolyte Interphase of Lithium Metal Batteries through Ultrathin Electrically Insulating Fluoropolymers** — score: 0.500 The article discusses a novel approach to enhance the stability of lithium metal batteries by using ultrathin electrically insulating fluoropolymers to suppress the formation of metallic lithium within the solid electrolyte interphase. This method aims to improve battery performance and longevity by preventing lithium dendrite growth, thereby addressing a critical challenge in battery technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **High-Performance Elastomeric Lithium Metal Anodes Enabled by a Lithiophilic Monolayer-Assembled Nano-Crumpled Micro-Concave Architecture (Adv. Energy Mater. 27/2026)** — score: 0.500 The article introduces a novel elastomeric lithium metal anode featuring a unique nano-crowned micro-concave architecture enhanced with lithiophilic amine groups. This innovative design significantly reduces dendrite formation and improves cycling stability, suggesting that such elastomeric anodes could greatly enhance the performance and safety of lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Engineering Thermally Resilient and Kinetically Active Reversible Protonic Ceramic**

**Cells via Interfacial Design** — score: 0.500 The article presents the development of a reversible protonic ceramic electrochemical cell (R-PCEC) incorporating an innovative interfacial engineering strategy and optimized electrolyte fabrication, which significantly enhances both the thermal stability and kinetic performance. Utilizing a combined perovskite nanoparticle layer and modified sintering techniques, the R-PCEC achieves impressive metrics, including a peak power density of  $1.112 \text{ W cm}^{-2}$  and sustained operational stability for 600 hours at  $600^\circ\text{C}$ , while effectively demonstrating improved electrode kinetics and resistance to thermal cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Ligand-confined zinc slurries enable durable flow-based energy storage** — score: 0.400 The study introduces a flowing zinc slurry battery that incorporates zinc nanoparticles within a stabilized conductive network using a ligand, addressing the degradation issues associated with traditional zinc electrodes. This innovative design allows for effective reversible  $\text{Zn}/\text{Zn}^{2+}$  conversion, leading to enhanced durability in energy storage applications. Journal: *Nature Energy*